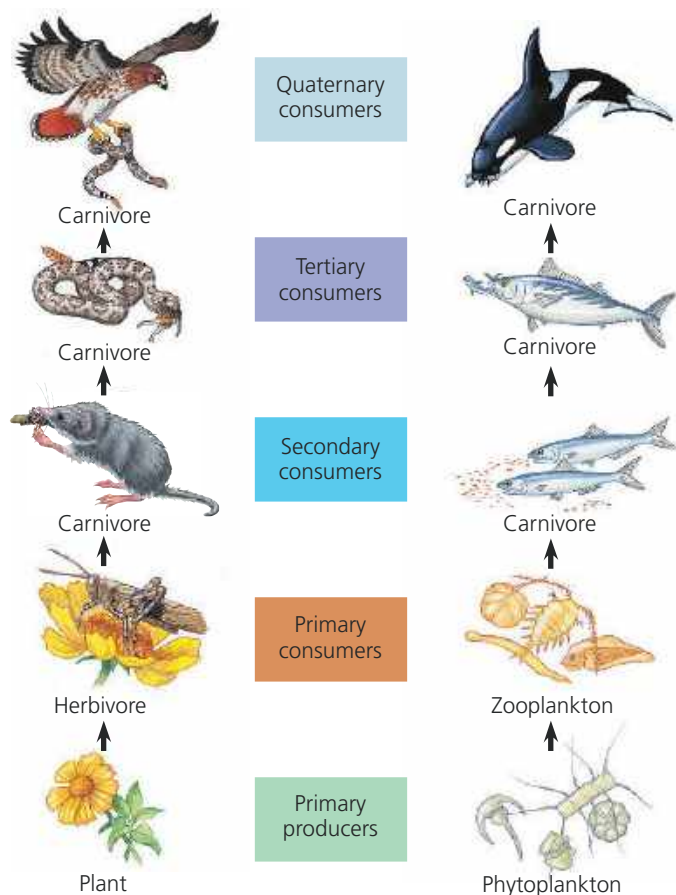


▼ **Figure 54.16 Examples of terrestrial and marine food chains.** The arrows trace the transfer of the chemical energy in food through the trophic levels of a community when organisms feed on one another. Decomposers, which consume the remains of organisms from all trophic levels, are not shown here.



A terrestrial food chain

A marine food chain

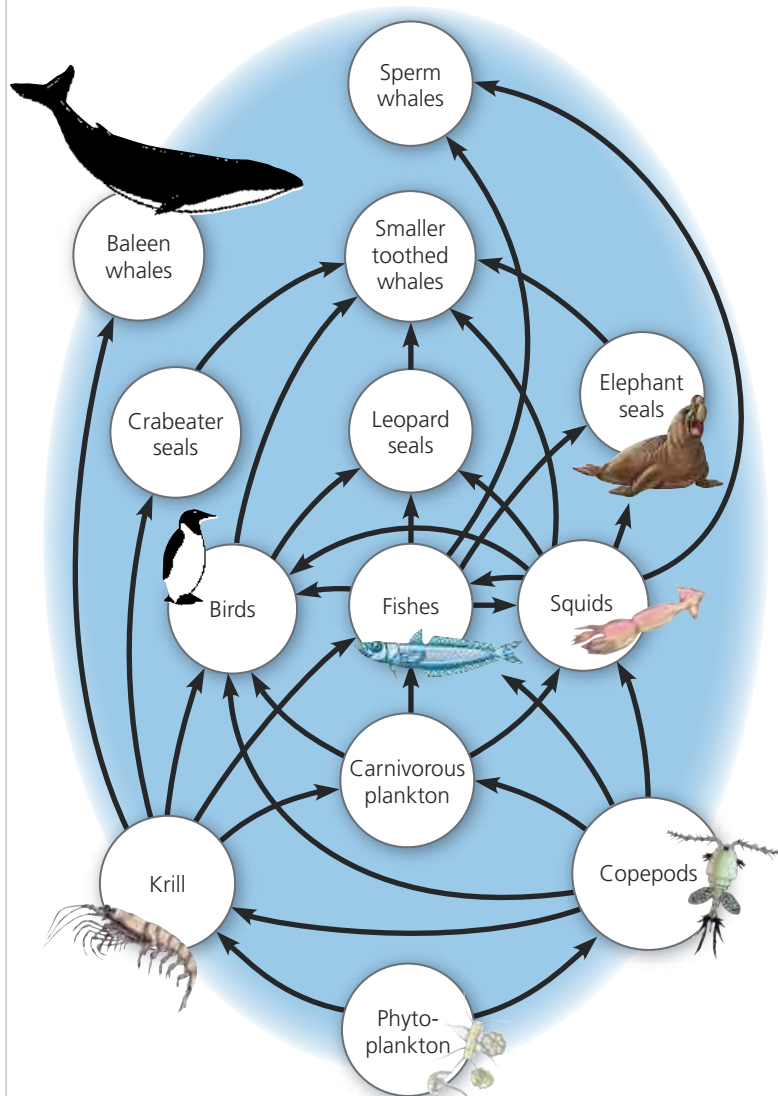
VISUAL SKILLS Suppose the abundance of carnivores that eat zooplankton increased greatly. Use this diagram to infer how that might affect phytoplankton abundance.

quaternary consumers) and eventually to decomposers is referred to as a **food chain** (Figure 54.16). The position an organism occupies in a food chain is called its **trophic level**.

Food Webs

A food chain is not an isolated unit, separate from other feeding relationships in a community. Instead, a group of food chains are linked together to form a **food web**. Ecologists diagram the trophic relationships of a community using arrows that link species according to who eats whom. In an Antarctic pelagic community, for example, the primary producers are phytoplankton, which serve as food for the dominant grazing zooplankton, especially krill and copepods, both of which are crustaceans (Figure 54.17). These zooplankton species are in turn eaten by various carnivores, including other plankton, penguins, seals, fishes, and baleen whales. Squids, which are carnivores that feed on fishes and

▼ **Figure 54.17 An Antarctic marine food web.** Arrows follow the transfer of the chemical energy in food from the producers (phytoplankton) up through the trophic levels. For simplicity, this diagram omits decomposers. At various times over the last two centuries, humans have also played a role in the Antarctic food web as consumers of fish, krill, and whales.



VISUAL SKILLS For each organism in this food web, indicate the number of other kinds of organisms that it eats. Which two organisms are both predator and prey for each other?

Mastering Biology Figure Walkthrough

zooplankton, are another important link in these food webs, as they are in turn eaten by seals and toothed whales.

How are food chains linked into food webs? A given species may weave into the web at more than one trophic level. In the food web shown in Figure 54.17, krill feed on phytoplankton as well as on other grazing zooplankton, such as copepods. Such “nonexclusive” consumers are also found in terrestrial communities. For instance, foxes are omnivores whose diet includes berries and other plant materials, herbivores such as mice, and other predators, such as weasels. Humans are among the most versatile of omnivores.